Decomposing Egocentric Spatial Relationships with Intermediacy and Imagery

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Abstract

Natural language communication about spatial relationships and orientations is a crucial aspect of human-robot interaction. This work continues efforts to represent natural language expressions about spatial relationships in qualitative non-standard coordinate systems while reaping the benefits of primitive decomposition. In this short paper, we propose cognitive representations of egocentric spatial relationships based on a "betweenness" spatial primitive and imagery-constructed region objects. We demonstrate how the scheme can represent left and right relationships between pairs of objects and "on the left/right" relationships between an object and an observer.

Introduction

Natural language communication about spatial relationships and orientations is a crucial aspect of representation systems for human-robot interaction and an enduring challenge to foundational models (Rahmanzadehgervi et al. 2024). Prior work has decomposed language meanings into complex combinations of conceptual primitives for aspects of mental models such as motion, containment, and theory of mind (Zhou, Duah, and Macbeth 2022; Macbeth, Zhang, and Badhan 2025). The work presented herein continues these efforts by representing natural language expressions about spatial relationships in qualitative non-standard coordinate systems with hopes of reaping the benefits of primitive decomposition (e.g., Zwicky 1973).

In this short paper, we propose cognitive representations of egocentric spatial relationships based on a "betweenness" spatial primitive and imagery-constructed region objects. We demonstrate how the scheme can represent left and right relationships between pairs of objects and "on the left/right" relationships between an object and an observer. The representational task and tests of our scheme draw inspiration from experimental psychology research on mental model representations of linguistic expressions in English about the spatial relationships between objects (Mani and Johnson-Laird 1982).

The following section introduces imaginary ground objects and representations of left and right relationships between pairs of objects. An included subsection introduces a third imaginary ground object and constructs representations for single objects and their general left and right orientations in relation to the observer's point of view. The remaining sections discuss challenges and ideas for future work.

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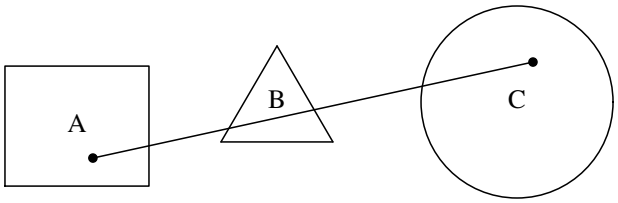


Figure 1: The `InBetween()` primitive is an intermediacy primitive for objects, shown here in two dimensions with objects of various shapes. `InBetween(A,B,C)` is irrespective of the centers of objects, and it holds if any point in `B` is on a line between any point in `A` and any point in `C`.

Decomposing Egocentric Spatial Relationships

We represent spatial orientations of objects as collections of relationships in which one object stands in between two other objects. The `InBetween()` primitive is an *intermediacy* primitive, shown in Figure 1 in two dimensions with objects of various shapes. `InBetween(A,B,C)` is defined to hold when there are points x in `A` and y in `C` such that at least one point on the line from x to y is in `B`. `InBetween()` is irrespective of the centers of objects, and alternatively, in mereological terms, it holds if any part of `B` is on the line of sight between any part of `A` and any part of `C`.

To represent egocentric left/right spatial relationships with `InBetween()`, we define *cognitive imagery-defined spatial regions* to be non-existent but conceived areas of space which are created and maintained cognitively and used in spatial cognition. We allow the operands of `InBetween()` to be real objects or cognitive imagery-defined regions.

Left Ground and Right Ground

We are inspired by recent work which decomposes "up", "down", "above", and "below" by introducing the ground or the center of the Earth as an operand of betweenness rela-

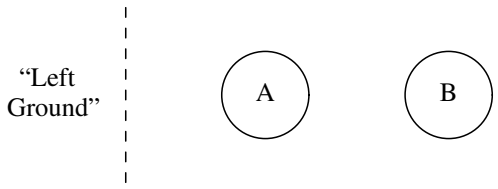


Figure 2: Natural language expressions such as "A is to the left of B" can be represented by asserting that A is in between B and `LeftGround`, an arbitrarily distant and arbitrarily large region of space to the left of the egocentric observer. Without loss of generality, we show real objects as equal-sized circles. The observer is not shown but is assumed to be facing the top of the page.

tions (Macbeth, Zhou, and Zhao 2023). We decompose "to the left of" and "to the right of" body-relative spatial relationships between pairs of objects using `InBetween()` and distant imagery-defined spatial regions which we call "ground" regions that are relative to the observer's positions and that represent the observer's frame of reference. In figures, without loss of generality, we show objects and observers as similarly-sized circles.

We name two imagery-defined spatial regions and then represent spatial relationships between objects in space by asserting intermediacy relationships that involve the real objects and these regions (Figure 2). We define `LeftGround` as an arbitrarily distant and arbitrarily large region of space to the left of the egocentric observer. Then we can define relationships where, from the observer's viewpoint, one object is to the left of another by representing that the object to the left is between the other object and `LeftGround`.

Natural language expressions such as "A is to the left of B" can be represented by asserting that A is in between B and `LeftGround`, or, slightly more formally

```
InBetween(LeftGround,A,B)
```

The equivalent language expression "B is to the right of A" and other equivalent language expressions can also be represented with this assertion.

We also define `RightGround` as an arbitrarily distant and arbitrarily large region of space to the right of the egocentric observer. Then we can define relationships where from the observer's viewpoint, one object is to the right of another by saying that the object to the right is between the other object and `RightGround`. Assertions such as "C is to the right of B" can be represented by asserting that C is in between B and `RightGround`, or, slightly more formally,

```
InBetween(B,C,RightGround)
```

We immediately recognize that, in this kind of representation, there are multiple sources of redundancy and non-uniqueness. We provide inexhaustive examples and delay addressing this issue until the Discussion section.

Center Ground

We define `CenterGround` as an imagery-defined region of space that extends arbitrarily far in front of and behind

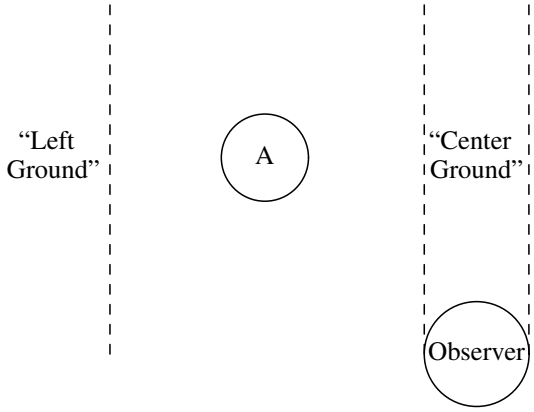


Figure 3: `CenterGround` is an imagery-defined region of space that extends arbitrarily far in front of and behind the observer and is roughly the width of the observer's body. Natural language expressions such as "`A` is on the/his/her/their/your left" can be represented as asserting that `A` is in between `LeftGround` and `CenterGround`. Observer is assumed to be facing the top of the page.

the observer and is roughly the width of the observer's body (Figure 3). This allows representations for objects individually and their general left and right orientations in relation to the observer's point of view by asserting that they are between `CenterGround` and `LeftGround` or in between `CenterGround` and `RightGround`. Expressions such as "A is on the/his/her/their/your left" can be represented as

```
InBetween(LeftGround,A,CenterGround)
```

while "C is on the/his/her/their/your right" can be represented as

```
InBetween(CenterGround,C,RightGround)
```

when there are no other objects in the vicinity.

A Complex Scene

Figure 4 shows a more complex scene in which the `InBetween()` primitive can be used simultaneously for both left-right relationships between objects and for individual objects' left-right positions in relation to observer orientation. In this scene, objects `A` and `B` are both "on the left" in relation to the observer. This relationship can be represented with `InBetween()` primitive relations with `LeftGround` and `CenterGround`.

```
InBetween(LeftGround,A,CenterGround)
InBetween(LeftGround,B,CenterGround)
```

Object `C` is "on the right" with respect to the observer, which can be represented with an `InBetween()` primitive relation putting `C` in between `CenterGround` and `RightGround`.

```
InBetween(CenterGround,C,RightGround)
```

At the same time, `A` is "to the left" of `B`, which can be represented with an `InBetween()` primitive relation placing

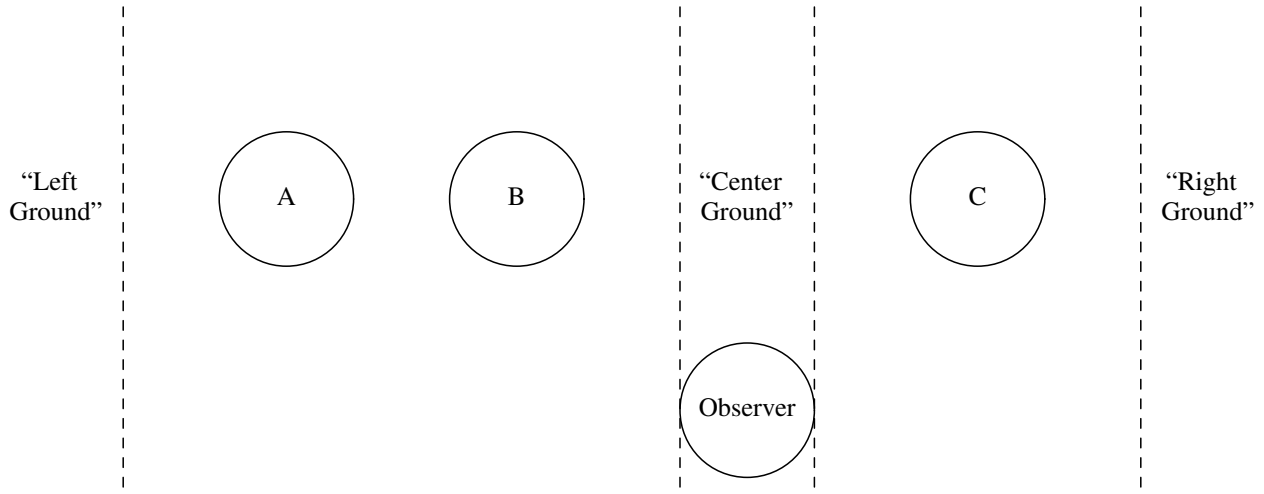


Figure 4: A complex scene featuring multiple real objects as well as `LeftGround`, `RightGround`, and `CenterGround`. `A` and `B` are both "on the left" in relation to the observer and object `C` is "on the right". `A` is "to the left" of `B` and both `A` and `B` are to the left of `C`. All of these relationships can be represented with `InBetween()` primitive relations involving real and imagery-defined objects and regions.

`A` between `B` and `LeftGround`. Both `A` and `B` are to the left of `C`, and of the many ways this can be represented, we can use `InBetween()` primitive relations placing `A` between `C` and `LeftGround` and `B` between `C` and `LeftGround`:

```
InBetween(LeftGround,A,B)
InBetween(LeftGround,A,C)
InBetween(LeftGround,B,C)
```

Of the many natural language "story" paraphrases which could be used to describe this scene, one is

> "B and A are on the left and B is to the right of A. C is on the right."

Discussion

We recognize immediately that the representation is non-unique in a number of ways. First, the `InBetween()` primitive is symmetric, so `InBetween(A,B,C)` implies `InBetween(C,B,A)`. This redundancy is an artifact of the conventional formal syntax used to define the relation and could be easily remedied in any implementation.

Also with both `LeftGround` and `RightGround` available, either could be used to represent "to the left of" and "to the right of" relationships. A scene like the one in Figure 2 could just as easily have been represented as

```
InBetween(A,B,RightGround)
```

Or, given the symmetry of `InBetween()`, it could also be represented as

```
InBetween(RightGround,B,A)
```

For left/right relations between objects, a canonical form which uses either `LeftGround` or `RightGround` but not both may be possible.

There is an alternative form of representation of "on the left/right" involving containment. One could define the space between `LeftGround` and `CenterGround` as yet another imagery-defined region with respect to the observer. Then, rather than a betweenness primitive, containment in this region could be used to represent that a particular object is "on the left". But this seems to be unnecessary effort, given that the region is defined with an `InBetween()` relation, as it is the space between two other regions.

Related Work

Many qualitative spatial relationship systems in robotics represent left-right relationships (e.g., Kunze, Doreswamy, and Hawes 2014; Kuipers 2000). The current work is unique in its attempts to decompose these relationships into combinations of even simpler primitives. *Betweenness* or intermediacy is the fundamental concept in ordered geometry (Coxeter 1969).

In the process of building a Conceptual Dependency parser for language about process paragraphs, Zhou, Duah, and Macbeth (2022) introduced the idea of representing "above" and "below" with spatial "maps" and an object representing the center of the earth. Macbeth, Zhou, and Zhao (2023) proposed a similar method for representing spatial orientation relationships such as "facing", "upside down", and "right-side up" in combinations of partonomy and intermediacy primitives, again using the center of the earth as an object or imagery-defined region. Another recent and related thread of work tested and explored primitive decomposition representations of theory of mind (Macbeth, Zhang, and Badhan 2025).

Conclusion

In this paper, we proposed cognitive representations of body-relative spatial relationships based on the `InBetween()` spatial primitive and imagery-constructed region objects and showed how left and right relationships between pairs of objects and "on the left/right" relationships

between an object and an observer could be represented. In future investigations we will test the representation by reproducing well-known experimental results on egocentric spatial relationships and mental models from experimental psychology (Mani and Johnson-Laird 1982), which will require betweenness-based decomposed representations of "in front of" and "behind" to represent the more complex scenes. We plan to use both determinate and incomplete scene descriptions as tests of spatial reasoning and inference.

Additional future work will perform tests of how the representational scheme can support live robotic environments, especially in tasks requiring perspective-taking, where a scene represented in a particular observer's point of view can be transformed to the viewpoint of another observer.

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